

Deep Learning-Enabled Intelligent Refrigerator: Design and Implementation of a Speech Recognition System

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Abstract. To address the inherent trade-off between energy efficiency and recognition accuracy in resource-constrained embedded devices, this paper proposes a lightweight speech recognition system **based on Keyword Spotting (KWS)** designed for smart refrigerators. We develop a parameter-controllable model architecture consisting of a shallow convolutional neural network (CNN) and a fully connected (FC) network, optimized through structural regularization techniques such as L2 weight regularization and Dropout. To mitigate the risk of overfitting and improve model robustness, we implement a multi-dimensional acoustic data augmentation strategy—incorporating noise injection alongside temporal and frequency-domain transformations—to effectively simulate the complex acoustic environment of a kitchen. Experimental results demonstrate that this multi-dimensional augmentation is critical to breaking performance bottlenecks; within the same lightweight architecture, the model achieves a test accuracy exceeding 97% with minimal training loss in only 2,000 iterations, confirming the efficacy of our collaborative optimization approach in maintaining high performance under strict resource limitations.

CCS CONCEPTS • Computing methodologies • Machine learning • Machine learning algorithms • Regularization

Keywords: Lightweight Speech Recognition; Convolutional Neural Networks (CNN); Acoustic Data Augmentation; Embedded Systems

1 INTRODUCTION

The advancement of voice recognition technology, fueled by breakthroughs in artificial intelligence (AI), has raised user expectations beyond the traditional experience of electronic devices. Speech recognition (SR) has moved from the laboratory to consumer-level applications, becoming the core interaction method for smart homes. Intelligent refrigerator is a field that is combining practice of speech recognition these years. Currently, intelligent refrigerators are in the arduous process of development. Manufacturers like Samsung are continuously releasing their new models of refrigerators that are capable of speech recognition. According to market research reports, the global smart home appliance market continues to expand steadily, highlighting the growth potential of intelligent refrigeration systems (Statista, 2024) [1]. This shows the potential of smart refrigerator and necessity of applying deep learning to it. With the application of speech recognition, refrigerator is no longer just a container used to store foods and drinks, it empowers people with ability of making new meals, which can significantly enhance life quality of users. It also enhances the enjoyment in kitchen and actively promotes smart kitchens.

The main purpose of speech recognition is to let machines know what people are saying. As stated by Wang and his team (2020) [2], a commonly adopted framework for Internet-of-Things-based speech recognition is Speaker-Independent Automatic Speech Recognition (SI-ASR). A typical ASR system contains several parts. Acoustic model training is the first part and is the base of speech recognition. The acoustic model extracts distinguishing features from speech units and statistically models their acoustic characteristics (Wang et al., 2022) [2]. Prior to model training, voice preprocessing is conducted to improve signal quality. Voice Activity Detection (VAD) is employed to determine the start and end points of speech signals by eliminating silence and background noise. Pre-emphasis and framing techniques are subsequently applied to enhance signal clarity and stability.

What is after voice preprocessing is speech feature extraction. Sound signal is a kind of wave signal; thus, computer cannot understand the meaning of it. To transform sound signal into computer understandable signal, it's necessary to perform correlation transformation on the waveform. MFCC (Mel-Frequency Cepstral Coefficients) is the classic extraction method. It simulates human ears and converts speech signal into the feature vector of speech signal. Up to this point, data is prepared and ready for model training. Different

algorithms and networks are then trained with processed speech signals, and with optimizations on arguments, accuracy of speech recognition can be increased.

Despite the potential of voice-controlled interfaces, deploying speech recognition in smart refrigerators faces several persistent challenges. One of these challenges is about energy consuming of speech recognition system. Large-scale matrix multiplications and memory access operations dominate the energy cost of deep learning systems (Horowitz, 2014; Sze et al., 2017) [3,4]. However, smart refrigerator operates as embedded systems with limited computational power, memory capacity, and cooling capability, making continuous execution of full-scale ASR models impractical. Deep neural networks impose substantial computational and energy burdens during inference, which presents significant deployment challenges for resource-constrained devices (Lane et al., 2015) [5]. Existing algorithms about speech recognition focus on recognition accuracy rather than energy consuming. Thus, those algorithms have many parameters, and with complex connections between layers and layers and a large number of energy consuming is required to conduct a large scale of matrix computation. This shortage makes it hard for existing speech recognition models to be applied on terminal end like refrigerator. The main chip of smart refrigerator possesses limited computing power, memory resources and cooling ability, making it difficult to continuously support high-consuming inference work. Therefore, directly deploy unoptimized existing models will result in shortened device life, severe overheating and even affect system stability, which is unacceptable in consuming-level products. Han et al. (2015) [6] introduced a method called deep compression, which is a three-stage pipeline: pruning, trained quantization and Huffman coding. Their method reveals that compressing model scale can actively reduce energy usage. Clarke et al. (2025) [7] conducted an experiment to compare the energy consumption between Automatic Speech Recognition (ASR) and Keyword Spotting (KWS) models with different parameter scales. The result shows that with less parameters, per-inference energy dropped by 24.8% for ASR and 31.2% for KWS. The results indicate that reducing the number of model parameters leads to a noticeable decrease in per-inference energy consumption. This suggests that model size is an important factor influencing energy efficiency in speech recognition systems. What's more, researchers also find that the word error changed by no more than 0.2 absolute, which means smaller model has a similar performance as large one. Therefore, the key in achieving the large-scale and practical implementation of this technology in products is how to design a lightweight model that performs a high accuracy.

To directly address the issue of high energy consumption, we abandoned the complex and interconnected deep network with numerous parameters and instead adopted a meticulously designed lightweight convolutional neural network. This is consistent with the conclusion of Clarke et al. (2025) [7] that model size is a key factor determining energy consumption. By consciously controlling the model size, we significantly reduced the amount of matrix operations required for performing recognition tasks, thereby meeting the limitations on computing power, memory and power consumption imposed by embedded devices. We also introduce an efficient training and regularization strategies to maintain the performance of small models. Purely minimizing the scale of model will result to a loss on accuracy, therefore, targeted regularization can effectively reduce overfitting of small models and enhance generalization ability.

This paper studies the intelligent refrigerator voice recognition system based on deep learning. The introduction elaborates on the background of smart home and the application value of voice recognition in the human-computer interaction of refrigerators and analyzes the limitations of existing technologies and the necessity of research. The system design constructs a framework that uses convolutional neural networks for feature extraction and fully connected networks for classification. Pooling layers are introduced to reduce dimensions, and regularization and Dropout strategies are adopted to enhance the generalization ability of the model. The experimental part conducts pre-processing of voice data, training with multiple sets of parameters, and testing to compare and verify the effectiveness of the model. Finally, the conclusion summarizes the rationality of the system design and experimental results and looks forward to future research directions in dataset optimization, accuracy improvement, and application scope expansion.

2 SYSTEM DESIGN

2.1 Introduction of system architecture

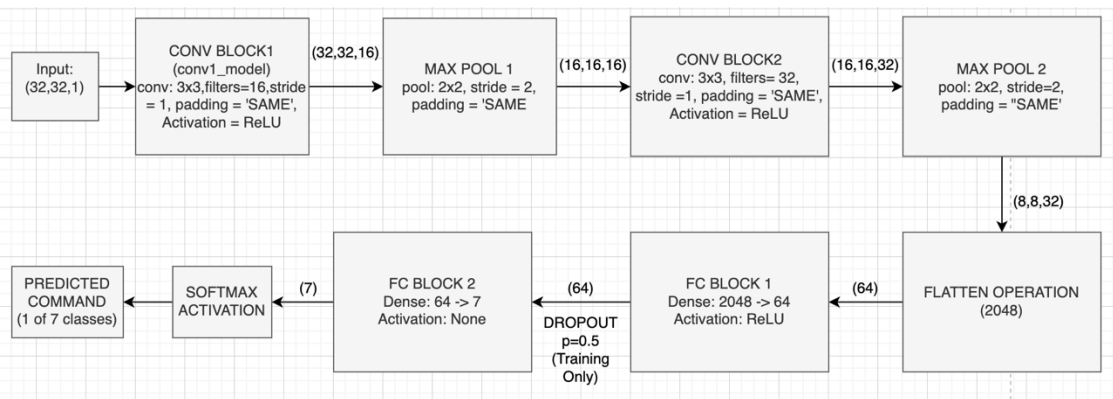
2.1.1 Overall architecture of the proposed system

This section introduces the overall architecture of the proposed deep learning-based intelligent refrigerator speech recognition system. Most common speech recognition models have about 5-30M parameters, however this system has about only 0.1M parameters. This system is mainly composed by three parts: a convolution neural network for feature extraction, pooling layers for dimensionality reduction and a fully connected neural network for classification.

As shown in Figure 1, inputs are preprocessed before being fed into the network. Then, the first convolution neural network with 2 convolution layer and 2 pooling layers will extract high-level patterns and deduct dimensions respectively. Finally, fully connected layers will finish the speech classification task.

The overall workflow of the system can be summarized as follows:

Speech Signal → Feature Extraction → CNN Layers → Pooling Layers → Fully Connected Layers → Output Prediction.



1Figure 1. Structure of the designed deep learning network

2.1.2 convolutional neural network (CNN)

The convolutional neural network serves as the core component of the proposed system. As the feature extraction engine of this system, CNN is responsible for learning discriminative time-frequency patterns from the original MFCC spectra. This model employs a shallow convolutional structure with only two convolutional layers that can strictly control model complexity and computational load [Kacur et al, 2022] [8].

The first convolutional layer (Conv1) uses 16 filters of size 3x3, and the second layer (Conv2) uses 32 filters of size 3x3. Small kernels effectively capture local spectral features while significantly reducing the number of parameters [Ruan et al., 2022] [9]. Each convolutional layer is followed by a ReLU activation function to introduce non-linear transformations. There is a max pooling layer after each convolution layer. The pooling window size is 2x2 and with a stride of 2. These pooling layers progressively reduces the spatial dimensions of feature maps, significantly decreasing the parameters and computations required for subsequent layers. They make the model more robust to minor temporal shifts in the input speech features. At the same time, they can control overfitting by taking the maximum value in local regions.

This convolution-pooling structure design forms an efficient hierarchical feature abstraction process, achieving the transformation from low-level acoustic features to high-level semantic features with minimal network depth.

2.1.3 fully connected neural network (FC)

The fully connected neural network serves as the classification decision engine. The classifier consists of two fully connected layers. The first fully connected layer (FC1) contains 128 neurons and uses the ReLU

activation function. The second fully connected layer (FC2) has a number of neurons to be recognized (e.g., 30 classes based on your dataset) where the number of neurons is same as voice command categories and uses the Softmax activation function to output the probability distribution for each class. A dropout layer is integrated after the first FC layer; this design is crucial for lightweight models. During training phase, dropout randomly abandon a subset of neurons, which forces models not to rely on any specific features or remember the training set. This effectively prevent model from overfitting then train data, ensuring that the model remains a high recognition accuracy.

2.1.4 pooling layers

Pooling is a key operation for achieving lightweight and efficient design in this model. The role of pooling extends beyond the dimensionality reduction and provision of translation invariance.

Through down sampling, the pooling layer halves the size of the feature maps. This directly reduces the input data volume for the next convolutional or fully connected layer. This significantly lowers the multiply-accumulate operations and memory access bandwidth required by subsequent layers, which are the main factors affecting the power consumption of embedded devices.

In summary, this architecture systematically constructs a model with less parameters and computational load through a shallow CNN, compact fully connected layers, and pooling operations. This design directly targets the energy consumption bottleneck discussed in Chapter 2, enabling the model to achieve high-efficiency, real-time voice command recognition on resource-constrained terminal devices like smart refrigerators.

2.2 Optimization of the system

To effectively decrease the scale of the model while ensuring its robustness and generalization ability in complex real-world scenarios, we applied to important optimization strategies: regularization and dropout. With these strategies, overfit issue of lightweight models is overcome effectively, and the overall performance of model is significantly increased.

2.2.1 weight regularization

Weight regularization aims to impose constrains on model parameters, preventing them from learning overly complex or extreme (i.e., excessively large) weights during the training process. With weight regularization, model will learn more smooth and universal feature mappings. Our system employs L2 regularization, it added a penalty term to the loss function that is proportional to the sum of the squares of all weights. The volume of our model is progressively controlled with appliance of L2 regularization, it pushes the network tend to use more but smaller weights when fitting the training data rather than relying on a few weights with extremely high values. This decreased model's sensitivity to noisy or specific accidental patterns in the training data. L2 regularization works effectively and fits the tasks like speech recognition well (Li et al., 2024) [10]. For our lightweight model, overfitting issue is outstanding and easy to happen. L2 regularization works as a structural constrain, guide the model to optimize in more generalized regions of the solution space without changing the structure of entire network. This is essential for keeping a steady recognition accuracy in a complex sound environment such as kitchen.

2.2.2 dropout random abandon

Dropout is a strategy that dynamically changing the network structure during the training phase and is widely regarded as one of the most effective regularization methods in deep learning. In our system, dropout is employed after the first FC layer. In training phase, this layer will randomly set the output neuron of the previous layer to 0 with a preset probability.

During each training iteration, dropout is equivalent to sampling a different "sub-network" on the original network. This mechanism prevents the complex co-adaptive relationships between neurons because no single neuron can rely on the existence of other specific neurons. This forces each neuron to independently learn robust features, thereby enhancing the model's generalization ability. In the testing or inference stage, all neurons are retained, but their output values are scaled according to the dropout probability to maintain

consistency in expected values. Dropout is a simple and effective way to prevent model from overfitting (Srivastava et al, 2014) [11].

Overall, L2 regularization and dropout strategy is not a simple technological stacking. They work together to construct this Collaborative Optimization Framework for Lightweight Recognition System. L2 regularization imposes global constraints on parameter values, while Dropout performs dynamic perturbations in the network structure. Together, these two methods ensure that the model learns the essential speech command features during training, rather than the specific attributes or noise of the training set. This optimized combination is the foundation for the model to achieve both high energy efficiency and high reliability simultaneously on resource-constrained terminal devices.

3 SYSTEM EXPERIMENT

This section provides a detailed account of the experimental setup, data preparation process, and model training and evaluation results of this study. By conducting multiple sets of control experiments, we systematically explored the impact of different hyperparameters on the model performance and determined the optimal configuration.

3.1 data preparation and preprocess

The quality of experiment data would directly infect the performance of the model. The core contribution of this study lies in the fact that, through a strategic evolution from "sliding window enhancement" to "multi-dimensional acoustic data enhancement", coupled with the precise implementation of the code, we successfully overcame the generalization bottleneck of lightweight models under limited original data.

First, after original data undergoes standard processing such as pre-emphasis, frame division, and windowing, its Mel-Frequency Cepstral Coefficients (MFCC) are extracted as the core acoustic features. In this experiment, the coefficient dimension was set to 16, while retaining sufficient acoustic information, this effectively controlled the dimension of the feature vector, laying a foundation for the subsequent design of a lightweight model.

To enhance the generalization ability of the model and relieve the overfitting problem, our research initially applied the classic data augmentation technique based on sliding windows. By taking different offset on the time axis of the original speech sample, sliding windows generates multiple variant samples with subtle differences, therefore, it expands the effective size of the training set without collecting new data (Sriram et al., 2020) [12]. This enhancement strategy simulates the minor temporal variations in voice commands, forcing the model to learn more fundamental acoustic patterns rather than fixed time-aligned features. However, this type of method is essentially data augmentation, which involves transforming the existing data to increase diversity. The generated data variations are still confined within the distribution range of the original data (Kline, 2025) [13]. To overcome this limitation, for a more significant improvement, a more advanced data enhancement strategy was introduced to our project.

Unlike traditional enhancement, data augmentation aims to generate new and diverse synthetic data samples and it's a series of semantic-aware transformations applied to the MFCC feature space. Its core lies in the controlled perturbation of speech features from multiple physical dimensions, including:

Noise injection: Add controllable-intensity white noise or pink noise at the feature level. White noise simulates broadband random interference, while pink noise (whose spectral energy decreases as frequency increases) better conforms to the statistical characteristics of many real-world environmental noises (such as wind sounds, equipment buzzing sounds). This directly addresses the challenge of background noise in the kitchen environment. Noise injection is widely regarded as a key technology for enhancing the model's robustness to input perturbations and domain shifts (Erichson et al., 2022) [14].

Time domain distortion (time_stretch, time_shift): time_stretch uses linear interpolation to simulate the natural changes in the speaker's speaking speed; time_shift, on the other hand, simulates the temporal shift of the speech signal at the feature level. These two transformations enhance the model's robustness to speech rhythm and temporal changes. Studies have shown that time stretching and pitch shifting are effective enhancement methods for improving the performance of speech recognition models (Morrison et al., 2021 & Laaidi et al., 2026) [15,16].

Frequency shift: Perform cyclic offsetting in the Mel frequency domain to simulate the overall spectral shift caused by differences in vocal tract length (such as between adults and children) or minor changes in the frequency response of the recording equipment.

Volume scale and dynamic range compression: simulating the effects of different recording devices preprocessing algorithms by simulating the variation of recording volume gain and altering the dynamic range of characteristics through nonlinear mapping.

The result of our experiment also supports that data enhancement with multiple dimensions has a better result on the recognition accuracy.

3.2 result of experiment

3.2.1 sliding window enhancement result

Firstly, we used the enhanced dataset obtained through sliding window and conducted a grid search on key hyperparameters such as Dropout rate, base learning rate, and learning rate decay rate. Table 1 presents some representative experimental results.

Table 1. Sliding Window Enhanced Result

dropout	base learning rate	decay rate	iteration step	train accuracy (%)	train loss	test accuracy (%)
0.9	0.001	0.99	2000	41.24	1.69	50.45
0.7	0.001	0.99	5000	57.84	1.23	43.34
0.5	0.001	0.99	5000	57.13	1.28	43.28
0.7	0.0005	0.99	5000	57.84	1.32	45.46
0.5	0.0005	0.95	6000	68.62	1.02	45.74
0.5	0.0005	0.9	8000	69.60	0.98	46.13

As shown by table 1, despite numerous attempts at varying the hyperparameters, the best accuracy rate of the model on the test set has consistently been confined within the range of 45% - 50%. This indicates that under the sliding window enhancement strategy, the model's performance has reached a difficult-to-overcome bottleneck. Most experimental combinations exhibit a significant "training-test" performance gap. For instance, in the configuration that achieved the test accuracy of 45.74%, the training accuracy was as high as 68.62%, and the training loss was as low as 1.02. This indicates that the model has learned very well on the training set, but it cannot generalize its ability to the test set. This verifies that the sliding window enhancement can only provide limited diversity and cannot enable the model to learn robust features for dealing with unseen changes.

The result of this experiment stage indicates that purely rely on slide windows for data enhancement cannot satisfy the high recognition accuracy demand of smart refrigerators that work in the complex kitchen environment.

3.2.2 multi-dimensional data reinforcement result

To overcome the bottlenecks, based on the optimal hyperparameter direction determined in the first stage, we implemented a decisive change: 1) Replace the sliding window enhancement with the multi-dimensional acoustic data enhancement strategy implemented by data_augmentation.py; 2) Increase the number of model parameters to 50,000 to match a more diverse data distribution. The results are shown in Table 2 and Figure 3, and the performance has undergone a revolutionary change.

Table 2. Multi-dimensional Data Reinforced Training Data

dropout	base learning rate	decay rate	iteration step	train accuracy (%)	train loss	test accuracy (%)
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0.9	0.001	0.99	2000	99.9	0.4	96.94
0.7	0.001	0.99	2000	99.8	0.6	97.36
0.5	0.001	0.99	2000	98.9	1.4	96.79

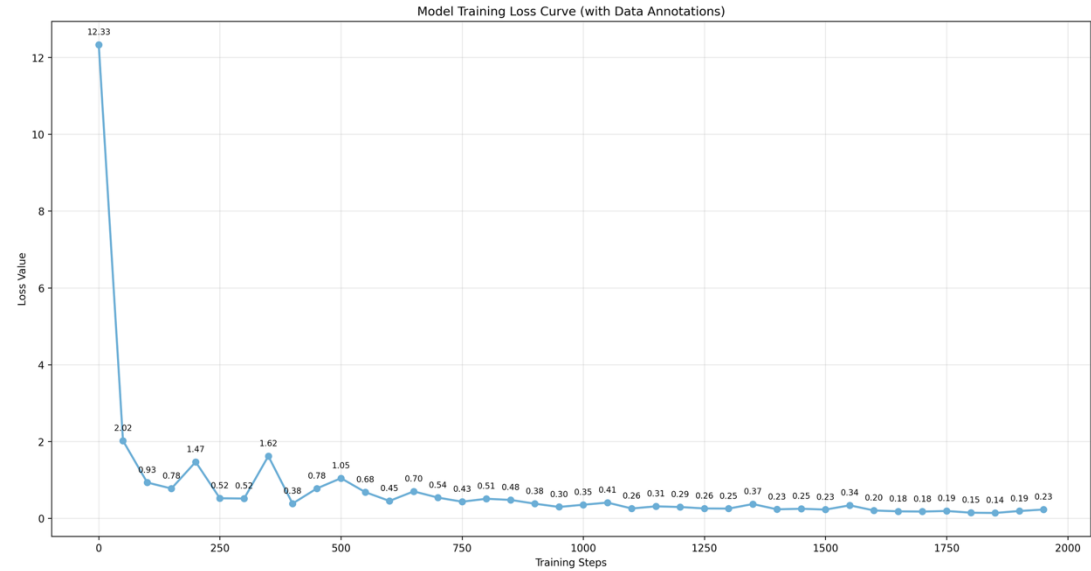


Figure 3. loss curve graph on the best accuracy situation

After adopting data reinforcement, with only 2000 training steps, the model's accuracy on the test set soared above 95%. Among them, the best performance was achieved when the dropout rate was 0.7, with a test accuracy of 97.36%, and the training accuracy and loss indicated that the model converged rapidly and well. By comparing Table 4.2.1 and Table 4.2.2, the most core conclusion can be drawn: Under the condition that the main model architecture remains unchanged and the hyperparameter range is similar, switching from sliding window enhancement to multi-dimensional data reinforcement led to a significant increase of over 50 percentage points in test accuracy (from approximately 46% to >97%). This directly proves that the essential improvement in data quality is the primary driving force for the performance breakthrough, fully confirming our discussion in the theoretical part about data reinforcement being superior to simple data augmentation. The model after data reinforcement exhibits ideal characteristics: high training accuracy, low training loss, and high-test accuracy are all achieved simultaneously. This indicates that through semantic perception enhancement such as noise injection and time-frequency transformation, the synthesized data successfully simulates the complexity of the real world, enabling the model to learn highly generalized features during training, thereby fundamentally solving the overfitting problem.

Review on our experiment, under the condition of limited data diversity (enhanced sliding window), the performance of the lightweight model has a clear limit (approximately 46% test accuracy). By implementing a multi-dimensional data enhancement strategy that combines noise, time-domain, and frequency-domain transformations, this ceiling can be completely broken through, and the model performance can be improved to a practical level (with a test accuracy rate of over 97%). This performance leap is not primarily driven by minor adjustments to the model's depth or more complex hyperparameter tuning, but rather by a fundamental upgrade in the data preparation paradigm. This strongly supports the core argument of this study: For speech recognition systems deployed in resource-constrained environments (such as smart refrigerators), investing in advanced data enhancement strategies is the most effective way to achieve highly robust and accurate models.

4 CONCLUSION

This study successfully designed and verified an efficient and highly accurate lightweight speech recognition solution for resource-constrained devices such as smart refrigerators. By combining a carefully designed shallow convolutional neural network with an innovative multi-dimensional acoustic data enhancement strategy, it effectively alleviated the prominent contradiction between computational complexity, memory usage, and model generalization ability in terminal deployment. Experiments show that the proposed multi-dimensional enhancement paradigm is the key to breaking through the performance bottleneck, enabling extremely lightweight models to achieve a test accuracy of over 97% with a limited number of iterations, significantly outperforming traditional data enhancement methods. This proves that the collaborative technical path of "advanced data enhancement" and "compact model architecture" has significant superiority and practical value in resource-constrained scenarios.

The future work mainly focuses on the following directions: Firstly, to deepen and optimize at the data level, collect diverse audio data that is more closely related to real user scenarios, and explore more advanced generative enhancement technologies; Secondly, to continuously simplify and adaptively improve the model, explore technologies such as knowledge distillation and neural architecture search to further compress the model, or introduce efficient lightweight attention mechanisms; Thirdly, to expand the application scope and robustness of the model, move from isolated word recognition to continuous speech recognition or voice wake-up tasks, and optimize its deployment and application on different hardware platforms and other smart home devices.

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